

Active Inference BookStream #001.012 ~ "Governing Continuous Transformation"

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thank you hello and welcome it's March 28 2023 we're here in active bookstream number 1.012 off to you blue great so we will be continuing our discussion of the book governing continuous transformation by Bhajan kesri and this actually is going to wrap up our coverage of this book we're not going to go into the interviews covered in the final section but we're going to just cover this concluding part and then wrap it up now and then in a few weeks we will rejoin for a discussion section or a discussion session so if you've been following along with the live streams to this point please get in touch with us so that you can come on and discuss some of the contents of the book with us and hopefully the author is available to join us so we are the active inference Institute of participatory online Institute that is communicating learning and practicing applied active inference you can reach us at social media at all of these links this is also a recorded and an archived live stream so please provide us with feedback so that we can improve on our work all backgrounds and perspectives are welcome here we will be attempting to follow a good video etiquette for live streams um again there's many ways to participate and collaborate with the active Advanced Institute so if you are interested in active inference and I'm not yet collaborating with us just get in touch active inference.org is our website and there's a link to fill out a form if you want to contribute in other ways to The Institute besides joining us on a live stream all right so the goal of this bookstream is to conclude our discussions of this book so we are going to cover chapters um 12 and 13. I would skip to the road back so we're going to come to chapters 12 and 13 in this part three conclusion of the book so chapter 12 is the future research agenda and chapter 13 is outperformance and survival as a matter of self-organization so we can just quickly introduce ourselves before jumping into the contents I'm blue I'm an independent research consultant in New Mexico and I'll pass it off to Tyler I'm Tyler I'm a protocol designer and dab researcher also in California and Daniel I'm a researcher in California did I say California foreign I'm having visions of the beach okay so chapter 12 has um a few different sections again this is

like future research directions discusses language and communication embodied cognition artificial intelligence board capabilities developmental paths and then chapter 13 is really just a provocative conclusion section um and reminds us of some of the things that we've learned in the book so far and also about performance and survival as a matter of like free energy minimization so we will jump right into the content uh Tyler do you want to read this first um part of the abstract for us sure Effigy is the product of multidisciplinary research project connecting the dots among strategy governance and Neuroscience research opening a new conversation and headlines introducing the free energy principles strategy management FEG redefines the Nexus structure and cognition ABV so attention-based view advances the dynamic capability View and from the managerial to the core level that establishes strategy as an inactive Bayesian inference challenge in terms of future research four areas should Advance the optimization of FEG one language and communication in particular explainability purpose prediction models and predictions to body cognition in terms of a deeper understanding around the neurobiology underlying leadership's World engagement as well as false inference thanks and I'll do this next part so the third area is artificial intelligence in particular the strategy application of AI powered machine algorithms and fourth board capabilities developmental paths in particular the board's ability to collectively develop meta capabilities free energy governance as a framework provides the fundamental building blocks that Empower a firm to implement free energy minimization as a self-organized principle structure top-down bottom-up connectivity cognition environmental enactment and capabilities Duality management within the fpg framework many vibrant strategy research topics lend themselves to be repurposed as a matter of optimizing free energy minimization more specifically survival rather than just performance is the critical dependent variable so getting into sorry I'm trying to navigate my board so getting into the first section language and communication um so I just pulled a couple of sentence sentences out of this section um Bijan says that the effectiveness of free energy governance is critically determined by transparency in communication specifically around the explainability of corporate purpose obtain the most possible human energy for productive effort um and I think like this is really something the communication overall is an area that I would like to discuss with Bijan further um and but he does really kind of clarify some of the questions that we've been having over the last couple weeks here in this section and even maybe the next section about um what specifically would like like what specifically needs to be communicated to all members of The Firm um and at all levels so he really does kind of clarify um he says if organization is in emergent and

communication as we believe then it is not a being but a becoming and that's a quote from Taylor and Matt every 2000 and then he also says language is not a tool but a force um and with this I think he um specifically is referring to like a force that holds the firm together like the um you know like contributes to the self-organizing purpose and sustainability of the firm um and really he does uh talk a lot about the purpose of the firm both in this chapter and the next as like everybody needs to know what is the point for the firm's existence or for the corporation's existence and so I think that that is one aspect um that might not necessarily be like a governance strategy or like something that's like internal like a method of operation it's like you know we think about like the top level Executives but um something that like why are we here what are we doing what purpose are we serving as a corporation as a company and I think that like that is like one reasonable um thing that can be communicated to team members at all levels I don't know if you guys have any thoughts here or yeah so one thing that was like not totally clear on about what Jean was getting at here was what it means to have explainable purpose because on one level like the FEG like ultimate purpose as I understand it is just like survival I'm not sure if he meant purpose in like some other sense of like I don't know we are improving our customers access to health care or something like that that style level of purpose and I wasn't quite clear what he meant and he didn't quite get that many examples so I'm not sure if you guys have any thoughts or a clear sensible he meant yeah very good question it does seem to be like there's uh competing Communications around what purpose is is purpose in this deflationary sense merely survival and persistence at which point the clear communication would just be around whatever we are we need to continue even if that involves change versus more sub purposes or topical performances which that might resemble standard non-survival oriented purposes like to deliver the most X in y yeah and it's I'm sorry so I have a question also if survival is the purpose of free energy governance or if free energy minimization is the purpose like I think about you know when a firm swallows another firm like you know in m a right like so you know but then the little firm is not the purpose is not to survive right like the purpose is to become um like I'm automated or or whatever subsumed by the larger Corporation sorry sorry if I got you offer interrupted your thoughts yeah and I well I guess something that I feel like has been not maybe I just haven't gotten it but not necessarily consistently communicated as like what their purpose is at the highest level organization or at sub levels so like in the same way as like a pancreas cell may not know like the meta purpose of like the pancreas I don't know anything about biology so I'm not mistaving something here like does it is it necessarily true that a individual team has to always keep in mind the larger

like meta purpose of this organization or is it really okay to have something that's more like relevant to it's like subunit organization um and that hasn't always been clear to me I'm still kind of struggling with that yeah I said I do think that like Jean um specifically like when he was talking about purpose and I think that it comes more in the next chapter but it might have been this chapter I don't know it's all blurring together just one big chapter but um he talks about like purpose being at like on the front lines right like the people that are on the front lines like are as much determining the purpose as they are communicating the purpose so it might be not a top-down directional communication method like or like it's it's a reciprocal like communicating the purpose of the organization is probably like a bi-directional information flow ideally anyway so like people at the top don't know what the people at the bottom think the purpose is and the people at the bottom don't know what people at the top think the purpose is like there's no unity and then the firm has a divided purpose I guess yeah I think that makes sense I mean my my take on what he's saying is like it's like this co-created maybe amorphous thing that you're not necessarily like pinning down and defining a thing on like a landing page on your website maybe something that's more of like a shared felt and sense thing um and so like that's what I think he's saying but like he doesn't explain it that way so I'm not totally sure yeah it's so these are good like for sure okay unless there's anything else we'll go to embodied cognition um where he says Brain body and environment together form one integrated cognitive embodied inactive extended system recasting the phenomena of false inference and institutional psychosis as a matter of impaired prediction error minimization May highlight how critically firm performance is also a function of upper echelons embodied cognition as determined amongst other by psychological aspects as well as neurobiology so like in this section there was like a big like a whole bunch of like neuro um neuroscience like about dopamine and the reward system and predictionary minimization and obviously like a firm doesn't have like dopamine as a prediction error like satisfier or like reward but like they do like the whole firm is getting like a dopamine hit but like the individual members within that firm are getting that dopamine um I guess when their predictions are are correct so um yeah this was kind of like a it was like a big like stretch for me to jump into like I guess like the embodied cognition of the firm is the embodied cognition of the upper echelon but shouldn't it be like the embodied cognition of all of the firm members or how do you get that like bi-directional integration of information without like the bottom mattering just as much as the top here I don't know this was like fuzzy and weird for me well so I mean I think like at one level organization it's you know like all each member's individual like dopamine receptors

and everything right but then at the next level like you can abstract that away and say maybe it's like the response to uh financing like maybe having more access to like Venture Capital funding for example is a way of like motivating action for an organization or um visitor growth on the landing page it's just like the signal like hey like you're going in the right direction like double down here and keep keep moving or keep exploring so that's maybe how I would think of it at like an organizational level in a way that maybe feels kind of like at a different level than just the sum of the dopamine receptors of all of its members yeah like hearing you on the gamification metrics for the organization like hits on landing page for a number of subscribers or the number of viewers or the number of Twitter followers um all of those things that we know that are like rewarding us personally with the I mean like why I guess like that that's like like an easy bridge to cross regarding like dopamine and institutional success or prediction or minimization I think well what I mean by that is a little bit differently it's not like yeah like on an individual level like if you see more hits on your landing page like that's gonna have a dopamine response with you with you as an individual but I think some of these things like for example like press coverage or giving Venture Capital funding for example like those produce systemic changes in an organization that aren't just merely about the dopamine response in the individual and that's something that I feel like is like a higher level change that is like harder to explain um and I'm struggling in terms of the words from it but just like when you get like a lot of funding into an organization or a startup or you start getting traction on something like it produces a kind of like social contagion I don't know if that's the right word is more than just the sum of its parts of dopamine changes if it's individuals cool um anything else here or we can move on to artificial intelligence so here John says free energy minimization is a machine logic in its own right empowering living as well as artifactual organisms to optimally resist entropy and like this is I think um like a shout out to like the many uh people who are working on using active inference in machine learning instead of reinforcement learning so it's kind of great and like he really talks about like the ethics of that like what happens when we like you know have like he talks about the singularity right like when machines become like smarter than us and he says um that effectively the challenge is moving from coding like programming with even with active inference right so coding to ethics like how are we going to ethically like program the machines so that um you know they're not like self-driving cars aren't running over you know people or or whatever right like so there's the ethics and then to um survival like like to human survival right like like why like um yeah if the machines get to equip whatever they don't need

us anymore like why should they follow the ethics that we've programmed into them I guess yeah this was a very interesting section first we could imagine a kind of trolley car problem with some organization and some people and eventually the kind of decidability that's going to have to come into play there and the other provocative aspect in this from coding to ethics to survival is we often hear you can't get an ought from an is so no amount of mechanistic definition is going to result in an ethical imperative can't get an Offerman is and that kind of describes this Coatings to ethics um difficulty you can't really get an ethic just from how something is encoded but in a way survival is being proposed as a post-ethical imperative a first principles we couldn't necessarily go from is to ought but we can go from is through ought back to is which is survival and so it kind of reincinuates a type of first principle ethic more like a work ethic than like a moral ethic but that opens up a lot of questions about what kinds of Ethics are of what status is under or after the survival imperative yeah it's interesting because like I think with active inferences or free energy governance specifically like there are two ways where I think it's interesting one level is like thinking about an organization as an organism was like independent of my Human Experience of that organism of like what makes organ like this organism survive and then I think the reason why I care about it the most which is like my own Human Experience of like how meaning is created for me in an organization and I think ultimately like that's personally at least what I care about the most and then when I think about AI I'm like okay well like even if you have an embedded a on your system if it doesn't mean something personally to humans in that organization like I don't really care even if it makes the organization survive more like it more likely to survive um so it's just like interesting these two different takes of like okay even if it makes the organization more likely to survive like why do I care if that doesn't isn't meaningful to me as a human in that organization not sure if that makes sense um yeah so then um also says first order principles such as ai's free energy immunization will lay the foundation for governance models to increasingly self-organize um so I think that like the the like this is very meta it's like the um how Daniel is saying like we go from coding to ethics to survival I think like maybe ethics is like the fog between like coding and survival um where like free energy minimization um like is about persistence um and like can be used in coding and like will lay the foundation for governance models to increasingly self-organize so like self-organization based on free energy minimization I guess yeah the section was interesting and I I do think that there's uh like John is super excited about Ai and uh like we got a long way to go I mean we'll get there it's not impossible it's just gonna take some time I think I don't know any other thoughts here in

artificial intelligence I mean I have lots of other thoughts here but I think like getting to Sean's perspective will be really fun yeah I guess I'm just not I think the thing I really am not quite understanding here is the explainability piece and I think I was trying to mention this earlier of like if you have an AI That's like bubbling up insights about your organization saying like hey this is what's going on or like this is the right answer of what you should do in this situation but it isn't like understood by the people in that organization like does it even matter and is that what Bajan means by explainability right like even if AI gives you the right answer but it isn't like understood and integrated into the people not organization is that AI actually useful in our um and that's like the tension I've been grappling with here and I don't quite understand and he talks about a lot about explainability of predictions I'm not sure if that's exactly what he's talking about or not but good question for John um and then the last section for um future development is the board capabilities developmental paths so he says indeed the board is ultimately responsible for anticipating the firm's future capability constellation including that of the board itself to sustainably succeed with continuous strategic renewal however to avoid the cognitive traps related to individual board profiles that Layman and Wallin identified we must study the cognitive micro foundations of capability development at the board level to date this area represents a void in both strategy and governance research like so I thought that this was cool especially like with respect to the active inference Institute like as we are fleshing out our own board and like what like the board decides what this and future boards will be capable of doing like now and forever um it ends like they can always all this stuff can be changed with approval of the board right so I think that um I I liked this like capability constellation um phrase I thought that that was kind of neat because there's a lot of like there's a lot of little like like sub capabilities right within the board like there's committees and this and that and like everybody can do or take care of many different things or like what don't we want the board to do like it's the possibilities are pretty like endless and um expansive but the like what is purposeful and what is um what will gain like the most bang for the buck in terms of what should the board focus on that's kind of I don't know crazy so um and then he outlined a little bit of these cognitive traps um but I I didn't go too far into them from here so do you guys have anything here on board capabilities um oh so this so I did go actually a little bit into them so what capabilities are required at the level of the individual board member when the necessary firm capabilities are not yet developed and so as Bhajan is thinking about this he um goes to this Theory you which I had never heard of before and so he outlines three of the principles developed in this Theory you and it looks like that there's quite a few um but he um

talked about like what what are the capabilities required at the board member level when um when we're defining the firm capabilities like the board capabilities for now and forever and so these I really liked um so he says number nine practice principle number five is what it is practice deep listening and dialogue connect with your mind and heart wide open when connecting to other people and contacts open up all four channels of listening listen from what you know from what surprises you from the whole and from what you sense once to emerge the emerging whole and I really like this like four different ways of listening like it's something that we don't always think about we just listen right and we listen like in one way like we're just like I listened to I heard this but when you're like looking at listening in different aspects it's it's different right like listening from listening from what surprises you like from um you know can you listen to it four times and like pretend like you don't know anything pretend like how does this relate to what you already know I don't know I thought that this was really nice I don't know if you guys have like if you were excited about it as as I was as excited um or I couldn't read a collective sense making next or maybe Tyler wants to read to Collective sense making yeah sure and I was just gonna say I really love the convergence of like new Ag and corporate governance uh literature kind of converging in real energy governance it's wonderful um all right so number 10 Collective sense making so use social presencing theater and embodied knowing so whatever process you use to make sense of The Learning Journey what matters most is that you pay discipline attention to all the voices and manifestations of the field uh paying discipline attention means that you are intentionally holding back all inclinations to mix your own interpretations or Solutions just spend all that because you wanted the data of the field to speak for you the data this field cannot the field cannot speak to you if your mind is too busy okay so that was also cool and then um oh yeah but the data of the field cannot speak to you if your mind is too busy expressing opinions and proposing Solutions oh that cuts to the core um yeah sorry I guess I put that off and then whatever my slide making capabilities were for subparts um and then the last one the last principle is principle 22. I'm like no I'm like I want to read this Theory you like what is this all about I'm like totally interested in learning more about these principles um number 22 is create enabling infrastructures that allow the system to sense and see itself um and he quotes making a system of sentency itself can change the coordination Dynamics among the players in a field moving them toward ecosystem awareness as traditional governance models are increasingly uprooted in a discontinuous world there is not only a need for a new governance logic but correspondingly new board capabilities developmental paths

specifically Duality management as a dynamic board capability demands dedicated research um and that Duality management just to like remind everyone is like financial and strategic control so it's like both of those things not one or the other like they they both work together okay huh and then we can go into chapter 13 or if you guys have more thoughts about this is Theory you principles no um we can go into this uh chapter 13 Tyler do you want to read this um portion of the abstract for us this classical cause effect determinism upon which Paradigm of strategy management uh as well as governance research has been built is uprooted paparian academics Relentless pursuit of empirically establishing qualifying and then falsifying mechanistic cause and effect relationships is struggling to inspire the business Community which is itself we're doing too often with these self-inflicted limitations of its own of its very World models together Academia practice or challenge to reconnect organizing to that self-evancing dimension of the universe's uh information system fpg is as valid for business organizations as it is for political constructs or for that matter matter navigating geopolitical dislocations and death cycle disruptions read this last s from one top down bottom-up hierarchical disconnectivity and a top-down dominance of supervised content prescriptions to an urge to adapt to the outer World emitted incapacity for an active inference and three the non-existence of both a Timeless deep purpose as well as meta capabilities is naturally entropy-bound in the absence of a generative Bayesian model we become incapacitated indeed camping on seesaws is the categorical imperative of survival pursuing self-organization organization as a means not an end it is right here where the fundamental difference between traditional corporate governance and free energy governance models lie free energy governance sees through formal organization as organizing manifests itself as a self-organizing generative model operating as a whole within and without the firm embracing self-organization as a self-evancing universal information system um and so he makes like I just pulled out several claims that he made in this chapter and a couple like little summary uh sections because this is like the official summary he says together Academia and practice are challenged to reconnect organizing to that self-organizing or self-evidencing Dimension inherit inherent of the University's information system to purposefully create extend or modify the firm's resource base and that's a quote from Health at AI and he says to thrive and survive purposeful behavior is one of life's defining features but it is only possible if living systems operate as a whole it is only by managing information that the cell can impose order on the extreme complexity of its operations and therefore fulfill its ultimate purpose of staying alive and reproducing and this is equipped from nurse 2020 and like I just I don't know this is so

hard because um and like I'll always go back to like Mike Levin's beautiful explanation of the constraints imposed by the larger system on a smaller system because a cell's ultimate purpose is not sting Library producing in fact like if it's one cell sure like but in a multicellular organism some cells like the purpose of the cell is to die like I mean we all start off with like webbing between the fingers and that webbing is cells and then that webbing you know goes away like so that cell in like the construction of our existence and our morphology like its purpose is to die so um I guess that that's well like we were talking about M A earlier like the purpose of the smaller firm that gets subsumed by a larger firm is also to like die or become incorporated or to somehow integrate with the current larger firm and so these these constraints I I don't know so like the the nested hierarchical levels like each top level or each larger level constrains the function of the inner level and also that information um so anyway sorry I have to just like say that because it's otherwise it's not um staying alive and reproducing is not the purpose of cells sorry I don't know if you guys said that too um Bijan says free energy as an information theoretic rather than thermodynamic measure is uniquely positioned to measure and guide a firm's survivability as a self-organizing information as well as belief system um and he quotes this book from Lindsay 2021 Daniel have you read this I haven't read it but um it's like a Christian Memoir I guess uh he says Kristen recalls a thought he had as a teenager there must be a way of understanding everything by starting from nothing if I'm only allowed to start off with one point in the entire universe can I derive everything else I need from that in firsten's world the free energy principle is now the nearly nothing that can explain almost to everything um I like this quote from this book and it's like also like I've got to go read it now like I don't know if it's is it a Christian Memoir maybe it's just an interview or like a long article or something like that but I haven't read it but it's like um someone else like you know giving like a detail of Kristen's like life and ideas I guess I don't know if you guys have thoughts here no yes no so and so I just want to just call out uh we talked we just talked about like very early on like maybe it was like the fourth video we did talk about like if you're going what are we talking about like thermodynamic versus um information theoretic free energy like go back because we we really did get into that so if you're going what are they talking about we did talk about that quite a bit in one of the early um videos so now might be a good time to just like refresh your memory that we did detail that so if that's looking weird to you um Go Go Get It Go Get a refresher okay uh and then claim here in summary will dissonance as an error minimizing source for optimizing the accurate accuracy of posterior beliefs rather than explaining dissonance away is essential to firm survival and discontinuous

environments um and I liked this like informational dissonance part this was like a couple videos ago not that not that far in the past but um yeah yeah so this like brought up the initial communication question like does everybody need to know everything all of the time or um and also like your information getting updated with like the outside information to like resolve the dissonance anyway there's that do you get some thoughts here all right I'm gonna make somebody else read these because I'm tired of talking tired of hearing myself talk uh Tyler do you want to do this this is the building blocks of an active inference so this is a nice little summary uh recap of um what we've learned so far all right building blocks have been active inference so one is the four states that's active action sensory perception internal firm resources and external the hidden unknowns two one internal external markovian boundary the site where actions upon hidden States generate outcomes a feed in circular causality our perception and allow the Precision enhancing updating of power beliefs predictions three one first order principle free energy minimization that's a measure of minimizing equivocally error and uncertainty I.E the differences between predictions prior beliefs and outcomes posterior beliefs eventually everything becomes miserable um yeah and so Bichon makes this claim that once managers start approaching exploration in the form of an active inference linking in circular causality all four states conditioned by the Marco blanket and governance by free energy minimization we can straight away see that success and exploration is critically determined by the level of sophistication of a few things first prediction models a self-evidencing Eco Niche model that connects corporate Purpose with the very predictions The Firm stipulates second the bounded set of external states that the firm decides to act upon to bottom-up sample the world and test hypotheses with the least required computational resource investment and third belief updating resulting from the consequences posterior beliefs that the firm's actions generate and so this is also like a nice little summary section I thought um and this can he continues to say it is now self-evident to appreciate that action is Central to perception more importantly the firm's generative model and actions controlled ambiguity and uncertainty the firm's generative model is the first Eco Niche firm is more in control than it traditionally resists to believe the perception of uncertainty is after Observer I.E firm specific um and and I did like like that uh like real uncertainty versus the perception of uncertainty and when we're talking about like embodied enacted extended cognition it's all about like your perception of uncertainty versus actual uncertainty so like you know unless you can count cards you're uncertain of like if you're playing Blackjack you don't you don't know what your chances are of getting 21. but like if you can count cards right like just that's

just one example like you're less uncertain if it's a single deck Blackjack like you're less uncertain as to what are the possibilities as the game continues to unfold it's also interesting there with the um kind of Declaration of Independence like self-evident appreciation that reflects the real bright line between passive inference and active inference and it's like that claim does one agree or disagree and then if they did agree then they have already seen their own eye movement and decision to seek out this information as Central to their perception of that expression so it kind of serves as like a co-on of self-evidencing and then there are some Downstream applications for example one that he attacks many times throughout is that the external world is not uncertain there's not uncertainty in the Commodities Market there's an observer dependent uncertainty about external States and about action selection and active inference is it just is a generic family of strategies for jointly resolving uncertainty about external States and actions that bring alignment AKA provide pragmatic value so many of those implications of FEG are Downstream of the acceptance or acknowledgment that action is Central to perception very nice thank you um okay so he lists these requirements for out performance and survival Tyler do you want to read these for me yes all right one at a Timeless uh deep purpose to generative top-down bottom up prediction error minimization form of free energy minimizing inactive inference three principled values and aspirations determining Eco Niche prosperity and sustainability if all three elements are articulated and simple explainable language and communicated to all members within and Beyond the firm's boundaries then the firm's long-term process will be sustainably enhanced in form of self-organizing generative model the board of directors rather than executive leadership only must ensure that those elements come to life and more importantly stay alive thanks um and so sorry so this goes back to um like we were talking about purpose earlier like where does purpose come from who decides what the purpose is uh so this it was more talked about I think a lot in this chapter um and and we've also talked about it before through the book so this is a nice little wrap-up section um and then goes on to make the claim that free organization free organization okay I've said organizations so many times I think like if I were to count like I probably have said the word organization like a thousand times in this live stream Maybe not maybe 150. free energy governance is as valid for navigating geopolitical dislocations and debt cycle disruptions any organism that suffers from a one top-down bottom-up disconnectivity two top-down dominance of ideological and supervised content prescriptions three dominant logic to adapt to the outer World emitted self-inflicted incapacity for an active inference and not least four not an existence of a times purpose as well as Medicaid abilities

such as listening to what wants to emerge in the web of random interactions and observing
The Observer is naturally entropy-bound and incapacitated to navigate macro disruptions um
so he gives that like recipe for survival and outperformance um here so and in the last
requirements for our performance and survival so I think that those are just like the three those
are just like the inverse statements of each other like this is what we need this is what we can't
have um so that's a nice like way to wrap up like how free energy governance will contribute
to um the firm's outperformance and survival in the world competing with other firms and
such um do you guys have any thoughts here good um okay and then I just pulled this final
quote out and we kind of alluded to this earlier it says it does the uncertainty that human
humans face come from the inherent instability of the human landscape or from the
perceptions and belief systems that we have about the human environment and this is kind of
nice like what what is it that's unstable like where is the uncertainty is the uncertainty like in
our perception or is it in the world somewhere um I don't know this is a nice like thing to think
about and um in resolving that uncertainty through action I think like gets to kind of the core
of free energy governance here at the end of the book so um thanks for listening I don't know
if you guys have any final thoughts on any of the content presented here in these two chapters
or on the book overall or you know questions that are growing through your mind nothing like
that all right we're quick through the concluding sections and um we look forward to having
an open discussion with the author uh in a few weeks as well as other participants who have
been keeping up with the content so thanks guys thank you Tyler thank you blue farewell